The Media Connectivity Association




OVERVIEW AND UPDATES · BROADCAST AND MULTICAST

# Media delivery over 5G Broadcast

The 3GPP-based broadcast delivery system bridging OTT streaming and broadcasting

3GPP ETSI

NOVEMBER 2026 – www.5g-mag.com

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# About the project

The documents the 5G Broadcast project rests on

## ETSI TS 103 720

**Technical Specification**

5G Broadcast System for linear TV and radio: the functions, reference points and receiver categories.

## 3GPP TR 36.976

**Technical Report**

The overall description of LTE-based 5G broadcast.

## 3GPP TS 26.346

**Technical Specification**

MBMS protocols and codecs. The MBMS application programming interface and URL are in 3GPP TS 26.347.

FIND OUT MORE **Technology** **Specifications** **Reference Tools**

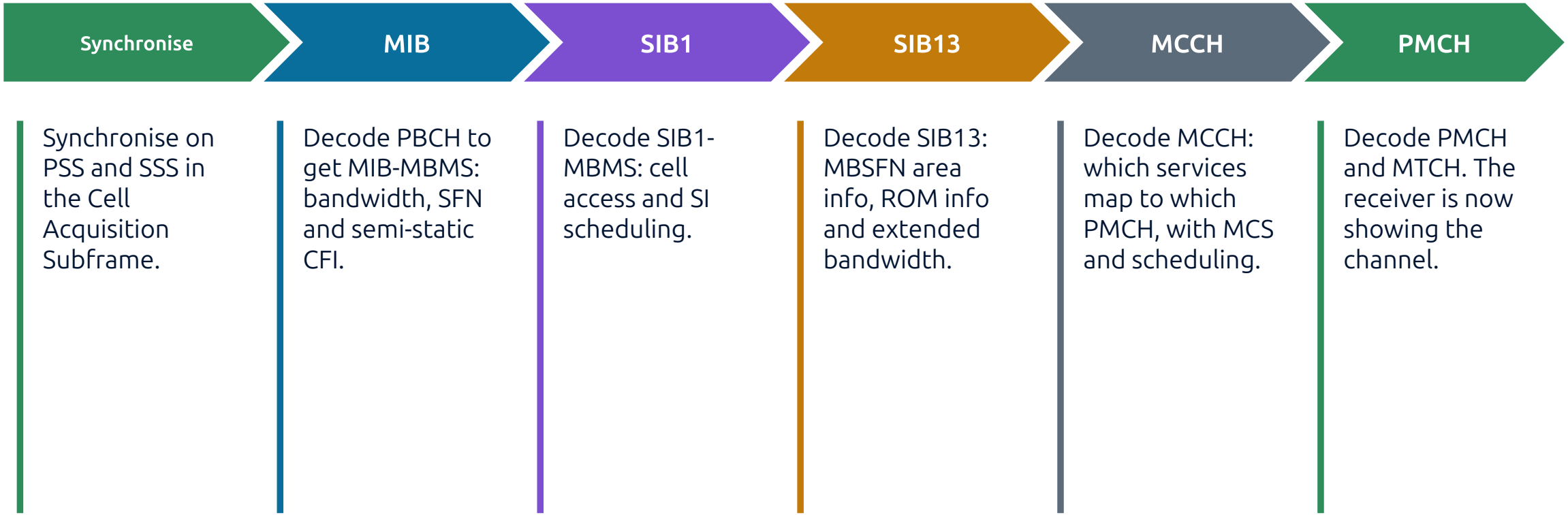


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# How a receiver tunes in

From nothing to a channel on screen, in the order the receiver acquires the signal




There is no SIM, no return channel and no prior connection to the network: the receiver bootstraps itself.

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# Features by release

What each 3GPP release added to LTE-based 5G Broadcast; ETSI TS 103 720 V1.2.1 profiles Releases 14 to 17

## Release 14 the enTV baseline

- MBMS-dedicated cell
- 1.25 kHz numerology
- Cell Acquisition Subframe
- 256-QAM on PMCH
- MIB-MBMS and SIB1-MBMS
- Shorter MCCH periods

## Release 16 LTE-based 5G Terrestrial Broadcast

- 2.5 kHz and 0.37 kHz numerologies
- PDCCH Format 4
- PBCH repetition
- MBSFN area info
- Semi-static CFI
- Receive-Only Mode

## Releases 17 and 18 bandwidths (17), UHF bands (18)

- 6, 7 and 8 MHz PMCH bandwidths (17)
- Band 107 (18)
- Band 108 (18)

## Release 19 PMCH Phase 2, CAS muting, new bands; not in V1.2.1

- Time interleaving
- Frequency interleaving
- PMCH cyclic shift
- Extended MCH scheduling periods
- CAS muting
- Bands 112 and 113

The UHF bands are receive-only (downlink-only). Release 20 (work plan of 28 September 2026): Band 246 (satellite) bandwidths, unverified.


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# How the specification grew

5G-MAG has maintained ETSI TS 103 720 since 2021, with changes agreed among members

## December 2020

First publication, with Release 16 as baseline: Free-to-Air and Receive-Only Mode services, single frequency networks over large distances, mobility up to 250 km/h, DASH, CMAF and HLS, IP-based services and file delivery.

## June 2023

Release 17 alignment: receiver requirements for network planning, 6, 7 and 8 MHz bandwidths and broadcast UHF spectrum, 5G Media Streaming over eMBMS, public warning and emergency alerts based on cell broadcast.

## In progress

The Release 18 update, started in 2024: DVB-I over 5G Broadcast, band support including Band 108, 5G Broadcast as a public warning improvement, concurrent 5G Broadcast and unicast, 3GPP Service URLs, a RAN-only receiver profile and low-latency distribution.

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# Receiver categories

Four categories in ETSI TS 103 720; the fourth, the Emergency Alerts Receiver, has no Feature Value

## Base receiver

**Feature value 27**

The minimum profile, aimed at fixed indoor reception. The 1.25 kHz numerology is mandatory; other subcarrier spacings are optional.

## Main receiver

**Feature value 28**

Supports all five numerologies: 15, 7.5, 2.5, 1.25 and 0.37 kHz. The target for mobile and portable devices.

## 5GMS receiver

**Feature value 30**

Meets all the receiver requirements plus the 5G Media Streaming requirements, for 5GMS over LTE-based 5G Broadcast.

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# The receiver chain

From the radio signal to a media player, as the Reference Tools receive it




The 5G Broadcast signal arrives as live I/Q data from an SDR, or from a prerecorded sample file.

Converts the signal into multicast IP packets on its output.

Decodes the FLUTE-delivered content from the multicast traffic and serves it to a media player.

Plays the stream in the browser with dash.js or hls.js.

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# Repositories: transmitter and receiver

Transmitter, modem, middleware and web interface, from the descriptions of the repositories

| REPOSITORY | WHAT IT IS | COVERS | LATEST RELEASE | |
|---|---|---|---|---|
| rt-mbms-tx | a standalone 5G Broadcast transmitter | Transmitter | ● v1.0.0 21 Jun 2023 | GitHub |
| rt-mbms-tx-for-qrd-and-crd | an MBMS-enabled eNodeB extended as a transmitter for QRD and CRD devices | Transmitter, QRD and CRD devices | ● ea-1.0.0 5 Sep 2024 | GitHub |
| rt-mbms-modem | converts an SDR or sample-file signal into multicast IP packets | Modem, SDR input, Sample files | ● v1.2.1 30 Jun 2022 | GitHub |
| rt-mbms-mw | decodes FLUTE content from the Modem and serves it from a web cache | Middleware, Web cache | ● v0.10.0 8 Dec 2022 | GitHub |
| rt-mbms-mw-android | an implementation of the MBMS Middleware for Android | Middleware, Android | ● 1.0.0 5 Sep 2024 | GitHub |
| rt-wui | a web interface that controls the Modem and Middleware and plays HLS | Web interface, HLS playback | ● v0.1.0 8 Dec 2022 | GitHub |

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● released ○ no release yet

More information at 5g-mag.com/tech/5g-broadcast

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# Repositories: core, applications and tools

The BM-SC, the Gateway, the Client, the applications and the tools, from the repository descriptions

| REPOSITORY | WHAT IT IS | COVERS | LATEST RELEASE | |
|---|---|---|---|---|
| rt-mbms-bmsc | the BM-SC: ingests content, announces it and delivers it with FLUTE | BM-SC | none yet | GitHub |
| rt-mbms-gw | takes content from the BM-SC and delivers it onto the broadcast bearer | MBMS Gateway | none yet | GitHub |
| rt-mbms-client | reads the announcement, reassembles FLUTE objects, republishes HLS or DASH | MBMS Client | none yet | GitHub |
| rt-mbms-application | a web app that plays the MBMS Client output and shows radio state | MBMS-Aware Application | none yet | GitHub |
| rt-mbms-application-provider | a provider portal that provisions services and sessions on the BM-SC | Application Provider | none yet | GitHub |
| rt-libflute | FLUTE version 1, which carries files over the broadcast bearer | FLUTE | 0.12.4 1 Sep 2026 | GitHub |
| rt-mbms-examples | examples that use other 5G-MAG repositories, such as FLUTE ffmpeg | Examples, FLUTE ffmpeg | 1.1.0 5 Nov 2024 | GitHub |

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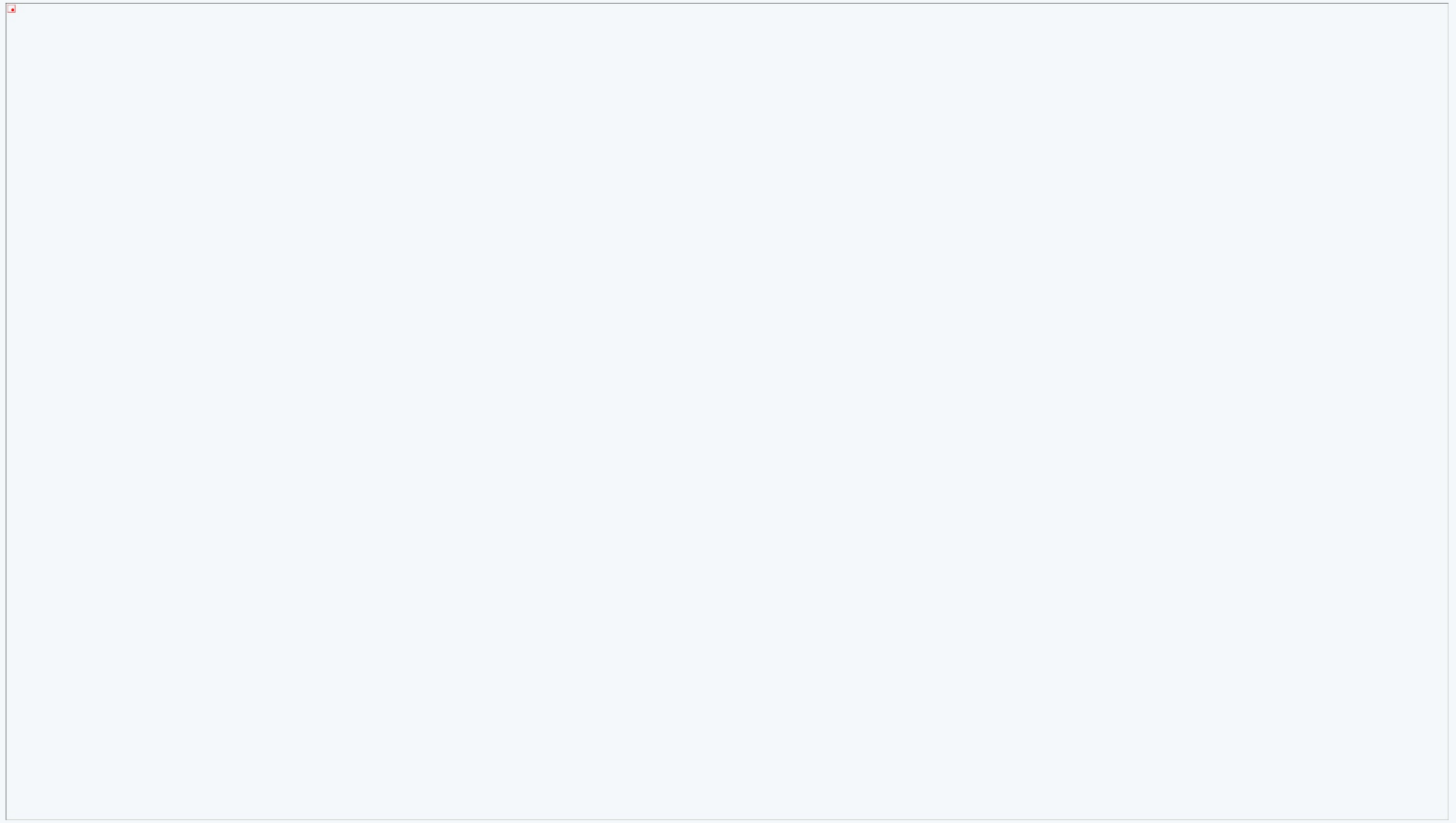
● released ○ no release yet

More information at 5g-mag.com/tech/5g-broadcast

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# Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



## PROVIDED BY THE REFERENCE TOOLS

1. **rt-mbms-application**
2. **rt-mbms-application-provider**
3. **rt-mbms-bmsc**
4. **rt-mbms-client**
   used by 1
5. **rt-mbms-modem**
   used by 1
6. **rt-mbms-tx**
   used by 3, 7
7. **rt-mbms-gw**
8. **rt-libflute**
   dependency, used by 3, 4
9. **rt-common-shared**
   auxiliary library, used by 4

Boxes without a badge are not mapped to a repository. The MME has no repository.

# See it and try it

A test setup with an SDR and phones, and the middleware app playing a live stream




**SDR and phones, test setup**




**Middleware app**

From the Reference Tools tutorials: seamless switching between broadcast and unicast on Android, and SDR playback.

WHERE WE STAND

# 5G Broadcast for TV and Radio Services

| BROUGHT TO STUDY | ON-GOING | OUTCOMES |
|---|---|---|
| Integration of DVB-I with 5G Broadcast | Release 19 transmitter and receiver | LTE-based 5G Broadcast transmitter |
| | Docker set-up | Seamless switching between 5G Broadcast and unicast |
| | Validator for the BM-SC, the RAN and the signal | Demonstration set-up for IBC 2024 |

## RELATION WITH SDOS AND INDUSTRY FORA

3GPP ETSI

**3GPP** Potential proposals on satellite backhauling under discussion

**ETSI** TS 103 720 (5G Broadcast): maintenance, no progress in this period

More information at **5g-mag.com/standards/5g-broadcast**

5G MAG The Media Connectivity Association

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